

👉 Lessons On Prototyping With Alberto Savoia

Alberto Savoia is the former Innovation Agitator and Engineering Director at Google, and Innovation Lecturer at Stanford. Founder of Pretotype Labs. He is also the author of a book that I loved, which is "The Right It: why so many fail and how to make sure yours succeed."

I took the chance to ask Alberto a few questions, and he was very kind to answer them all!

How did you get passionate about understanding why startups fail? What is your background?

Alberto Savoia: I never set out to study failure. Indeed, I thought that I was immune from failure. I was fortunate to join two successful startups, I was lucky to join Sun Microsystems, a company that became very successful, and I joined the company in the early days. I was also very lucky to join Google in the early days as its first engineering director and that became very successful. So I thought for a while that failure was something that affected other people, not me. And then, just as my hubris reached a peak, I raised 25 million dollars in funding to start a company that 5 years later failed, so I thought how is it possible that me, Alberto Savoia, would fail? What's happened? I thought I was destined for success. And as I write in the book (The Right It), the beast of failure bit me, and I decided to bite back. I decided, how is it possible that people who know what they're doing (because I thought I knew what I was doing) can fail?

So I set out to **study failure and to study ways to defeat failure and my mission these days is to help entrepreneurs and innovators all over the world to also fight failure and win**, and that's why I wrote the book.

Gennaro Cuofano: Failure rate seems to be the same across the board, even though we only read the stories of successful companies on magazines and media.

While studying failure you stumbled upon an idea, that would actually change your professional life. This is the idea of "pretotyping."

What is pretotyping, and how did you find out about it?

Alberto Savoia: You're right, the failure rate, whether it's new companies, whether it's startups or even small businesses, and even new products from existing and successful companies. The failure rate also affects the Googles and the Microsofts of this world. So it's not something that entrepreneurs and startups only fail at. Having said that, **the failure rate appears to be very consistent across markets, across time, it's almost like the Pareto Principle, the 80/20 rule. At least 80% and in some industries at least 90% or more of new products and new businesses tend to fail.** So when I set out to study failure, I thought, "well, why do they fail?" My first instinct was the people didn't build the product wrong. So maybe they were incompetent in designing it or engineering a new product, or executing if it's a new service or a new company. So I thought that the number one problem was incompetent execution.

But when I actually went and looked at the data, I found out that **the number one reason why most new businesses fail, why most new products fail, has very little to do with execution.** Of course, you want to execute it properly, **but the number one reason they fail is that people built what are called "The Wrong It."** What is The Wrong It? It's an idea that even if competently executed will fail in the market. So there's no amount of design brilliance, engineering excellence, or marketing fireworks that will save a product that is the Wrong It.

So I thought, okay, I want a product that is The Right It, which I define as a product that, if competently executed will succeed in the market. So I started looking at, how can I make sure that the product is The Right It? So I want to be sure that, if I build it correctly if I build it right, the product will succeed in the market.

And the first problem I stumbled onto is the way people conduct their market research. So, nobody wakes up in the morning and says, "well, I have a really bad idea for a new product or business and I'm going to waste two years of my life building it." No. Everybody thinks they have a great idea. And what they do is they go and present the idea to other people and they collect other people's thoughts and opinions. So I call this, **"market research done in Thoughtland."** So you ask people what they think of your idea and they tell you good or bad and then proceed accordingly.

I thought, well, this does not work right because people are really bad at predicting whether they would actually buy something, how often they would use it if they would use it at all, and how they would use it.

Thus, I thought there has to be a better, more reliable way of doing it. And that's when I stumbled into pretotyping. So **pretotyping is a word I made up, and as the name suggests, it is something that you build way before you build the product but also way before you build a prototype.** You don't actually have to build something to see if the market is interested in it. So that's what pretotyping stands for, it's something you do before prototyping.

How different is pretotyping from prototyping?

Alberto Savoia: The key difference between pretotyping and prototyping is the following: typically you build a prototype to make sure that what you are planning to build can actually be built to see if it will work, how it will work, how long the battery will last, etc. etc. So, prototypes are built to make sure that you can build something. Now, **my research shows that 99% of the time you can build it.** Most of the apps, say, in an app store do not fail because people cannot build them, they fail because people are not interested.

Pretotyping is designed to ask a very different question. It asks the question, "should we build it?" So, if we build it, will people buy it? So let's pick, in my book (The Right It), I have many many pretotyping techniques so we cannot cover them all but I'll give you a very simple one since we are talking about apps. So let's deal with the fact that most apps fail in the app store.

And the numbers are horrendous, like 95% of them don't make any money.

So let's say I have an idea for a new app that I can use to help me find the interesting books based on my personal preferences. I just came up with this in a flash. As an engineer, do you have any doubt that I can build this app?

Gennaro Cuofano: As an engineer, I know that you can build it, of course.

Alberto Savoia: Exactly, so the question you really need to ask is not can I build it, so it's kind of pointless to prototype because I know I can build it.

So all the uncertainty and all the risk lies not in building it but in the market wanting it. So how can I see if the market is interested in this app before I actually build it?

One of the techniques I teach, one of the pretotyping techniques I write in the book, is called the **"YouTube Pretotype," so instead of actually building the app with code, you can use PowerPoint or KeyNote or any other program to simulate what the app will do**, right? So you can make a little video, a little movie, that will show the potential users what the app actually does.

So you can have one screen where you write your name and you write the books that you like and then you click a button and it asks for recommendations and you see the app giving you a recommendation. You can do all of this without writing a single line of code. Anybody can do it if you can use PowerPoint or Keynote you can do that.

The way you do it is you create a video with that and then maybe you have a one or two page website, something very simple, where you describe what your app does; shows the video of the app; and then what you need to do is to show that to people and see if you can collect data that I like to call "skin in the game data."

So skin in the game data means not people telling you, "Oh yeah, if you build it, I will buy it," they need to give you something. So the smallest amount of steering the game somebody can give you is a valid email address with a clear understanding you will use that address to let them know if the app is ready. So you have this video, you either buy some ads or you put it on some forums, you say, "okay, here's an app I am trying to build, if you are interested, please give me your email address, and once I launch it, I will let you know."

The data you get there is enough, so if you show it to 1000 people and you get 3 email addresses, then it is probably going to be unlikely that there is a great demand for it. But at least you have some hard data, and if you don't find that enough people want it, then you do not bother building it.

What's the IBM Text-To-Speech and why it mattered so much to your idea of pretotyping?

Alberto Savoia: Many many years ago IBM thought "we want everyone to have personal computers," but there was no way (think about this is like 1980) that most people are going to learn how to use a keyboard. In those days who used a keyboard? Secretaries, programmers, and writers. So they thought, we need people to be able to operate the computer without using the keyboard, just by using speech to text into a microphone. Of course, they could not build the technology, they could not build the prototype for years because the technology was not there, computers were not fast enough.

But they thought, okay, maybe we want to make this investment, how do we actually make sure that people will want to use a microphone exclusively to interact with a computer? So they did a very clever thing, they brought people in the room, they gave them a microphone, and there was a screen in front of that microphone and told them, "Look, this a new way of running a computer, there's no keyboard, you just speak to it, and give it a shot and tell us what you think."

So, for example, somebody takes a microphone and dictate a letter to Mr. Jones. "Dear Mr. Jones, ..." and on the screen, they would see exactly their commands and the letters being written and the letter being sent.

Now, this was not possible, the technology wasn't there, so they could not have built a prototype. So what happened, in the room next door, instead of a computer that actually did all this processing, there was a human being, one of those people who know how to type very fast, got the input through the headphone, typed it on their keyboard, and so to the users it looked as if there was a working prototype, but there wasn't.

And the interesting thing, this is when I came up with the name pretotyping, originally I called it a pretendotype, because I thought, they haven't built something that actually works, they're pretending to have a prototype, so let's go with pretendotype. Then I shortened the name to pretotype.

And the beauty of this is they go from really valuable data, so, for example, they learn that people cannot talk to a computer for several hours a day because they get a sore throat. It's also very loud if everybody speaks at the computer, and also you cannot use it for confidential things because everybody overhears you.

They learned very quickly and they got very valuable data that, while Text-To-Speech may be interesting, they better focus on accepting the fact that people will have to learn a keyboard. And, surprisingly enough, 40 years later we've all learned how to use keyboards even though, arguably, they are not a very efficient way of using a computer.

Gennaro Cuofano: It's a brilliant story. As pointed out in this conversation failure is not something that just affects startups, but its something that affects also large companies like Google. Also, the rate of failure is not something that changes according to the stage of growth of a company. Large companies keep failing in some way, especially if they want to keep innovating.

I was looking a few weeks back at the Google Graveyard which is a set of products killed by Google because they actually failed miserably.

So you went back to Google to understand and to help people understanding how to embrace failure because its not easy also within companies like Google to say "You're going to work on this innovative product, which is not established" as most people want to work on successful products (like Gmail) rather than bet their careers on new and unproven products. So how did you manage to get your ideas accepted within Google? How did you manage to actually spread them?

How did you manage to help people embrace projects which were highly risky?

Alberto Savoia: There is an expression at Google which I write about in the book and that expression is **"data beats opinion."** So the way that you convince people at Google is not by waving your hands and being passionate about what you believe. I mean, of course, that helps, but the best way to make your case is with data. And fortunately, when it comes to failure, we have so much data, right, because most new ideas fail.

So I had no shortage of data and I was able to slowly, and initially convince various people at Google to pay attention to this mantra which is **"make sure you are building The Right It before you build it."**

The problem is not that new projects are going to fail, because it's much more exciting to work on a new project at Google that succeeds than one that is already successful. Gmail is safe but isn't it more fun to work on a brand new sexy project which also succeeds?

Thus, the way you beat failure, the way you overcome this reluctance to take risks or new projects is to make the point with data. So, we're going to build this project, but we're not going to ask you to spend two years to build it just to watch it fail.

Before we build it let's do a series of these experiments to give us very high confidence that if we invest two years to build it, it will then succeed. And that's how, initially, I introduced it at Google. And, by the way, I'm not the only one at Google who thinks this way.

As I say in the book, I didn't invent many of these methods, but I collected and I gave them a name. I'm kind of more of a curator and teacher and organizer of these techniques. So I want to be clear I don't want to take credit.

This way of acting at Google and many other companies based on data, not opinions, is something that a lot of people already do. I just organize it and explain it hopefully which you believe is in a clever, interesting, and memorable way in my book.

Gennaro Cuofano: And, as you said, it's very important that you get out of what you call "Thoughtland" and just to push your idea and make sure that you have data, valuable data, that is data that you collected that can help you understand whether you can go on with making the product.

Why then it matters to get out from Thoughtland and why you need to collect your own data?

Alberto Savoia: I want to emphasize that because that's one of the most important things I cover in the book. The idea of "Thoughtland" is a place where you create your ideas and then if you stay in Thoughtland it means you just go and ask people "what is your opinion" of your idea, which is dangerous, as we discussed, because people don't know what their opinion is.

They have a very hard time predicting. **So you want to get out of Thoughtland and you want to collect, this is very important, Your Own DAta.** Think about Elon Musk, right, one of the great inventors and innovators and entrepreneurs of recent times, when he thought about Tesla, if he looked at other people's data, all the other people and companies that tried to build electric cars and failed, he probably would have said "there's no way to succeed in this market."

Instead, he created his own data, he had his own vision for a very sexy and very fast electric car and he collected his own data. In fact, he prototyped it. I explain how in the book, but essentially **he got people to pay ahead of time, to give him money before building the car, because he wanted to some evidence that if he actually built it people would want it, and what better evidence than having cheques for \$5000 from people to demonstrate that they really really really want your electric car.**

What companion business books (to The Right It) do you suggest?

Alberto Savoia: I think there are two books that compliment with The Right. To be clear, I focus on the very very beginning. I call it the "big bang." From when you have an idea to make sure that you get the first step right.

So because the most important thing is to make sure you have the right idea. But two books come immediately to mind to compliment mine:

- The Lean Playbook by Dan Olsen: So whereas I focus very much on the first step, making sure that the idea is right, Dan gives you a playbook for the full steps.
- Unlearn by Barry O'Reilly: it is a very important book because you cannot take on new principles if you're not willing to unlearn previous principles.

The key to Unlearn is the following. In my context, the number one thing you have to unlearn is, do not do market research in "Thoughtland." **Instead of asking the question "if I build it will you buy it?" You need to flip it around 180 degrees and approach the market with the following research question: "If you buy it, we will build it."** Sort of like how Elon Musk did it. And pretotyping is a way to be able to ask that question and collect that market data.

Is there any business person nor entrepreneur you suggest following?

Alberto Savoia: Nobody on the top of my head. Everybody has their own, especially depending on what market you're in, you should focus on the market that you're in. And look at people that have been successful there. Again, **follow their inspiration but do not follow exactly in their footsteps because what worked for them isn't likely to work for you. Collect your own data about your own projects so you can do things your own way.**

What are you working next?

Alberto Savoia: Right now I'm busy trying to get the book into as many hands as possible because **my mission these days is to make sure to save entrepreneurs and innovators from failure.** I believe that innovators and entrepreneurs are the world's number one resource. They are the ones that are going to fix a lot of the problems that we create in the world. So my mission is pretty simple. Help them fight failure, the book is the first step, it's like your "Fighting Failure 101" and I'm just getting out and spreading the message and I appreciate your help in spreading this particular message.

Gennaro Cuofano: I'm very happy to help and of course suggesting this book to everyone and especially, "The Right It" is a great book and gave me the idea of how to prototype before I launch my next project so it has been great reading and it will be a great reading for many other people.

Thank you, thanks a lot for joining me for this conversation Alberto.

Alberto Savoia: Thank you, Gennaro, it's been a pleasure, good luck to you and good luck to all your listeners. May you always find The Right It!

Key takeaways

- The number one reason startups fail is not due to execution (which still matters) but due to the fact they build the "wrong it" or they start with the wrong idea in the first place
- Usually, the wrong idea comes from the fact that market researches are done in "Thoughtland." Thus they are based on biased data which is not good in telling whether an idea might work
- Data bits opinion is a mantra Googlers use to understand whether ideas make sense. Rather than rely on your own opinion, you need to make your thinking process as data-driven as possible. But not all data is good
- The good data is "Your Own Data" (YODA). Indeed, that is the data that enables you to make an informed decision on whether an idea has market potential
- YODA has to be skin in the game. In short, you want people to give you something, rather than just an opinion. In the real world, opinions don't matter. Also, a small action (as collecting an email) is a skin in the game data, compared to an opinion
- Pretotyping (pretending to have a working prototype) is a fast way to prove an idea. Rather than focus time, effort and money on a prototype that the market isn't wanting, with prototyping you can flip the questions "if I build it will you buy it?" and approach the market with the following research question: "If you buy it, we will build it."